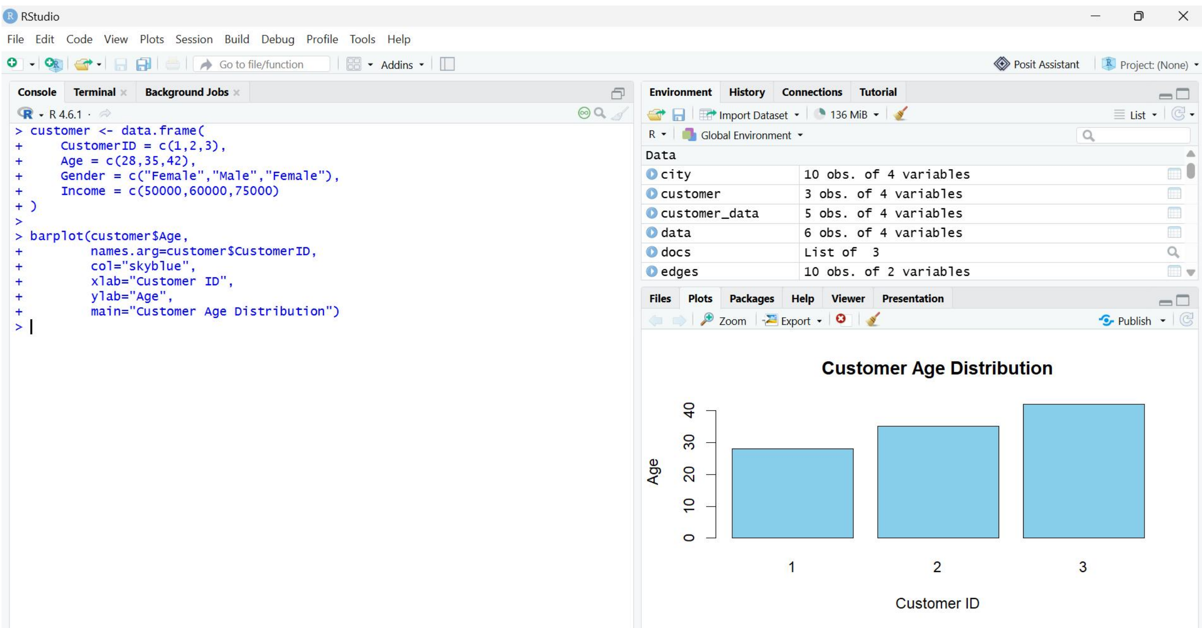
SET 7 – Customer Demographics Analysis

Q1. Bar Chart

```
customer <- data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)
barplot(customer$Age,
        names.arg=customer$CustomerID,
        col="skyblue",
        xlab="Customer ID",
        ylab="Age",
        main="Customer Age Distribution")
```



Q2. Pie Chart

```
customer <- data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
```

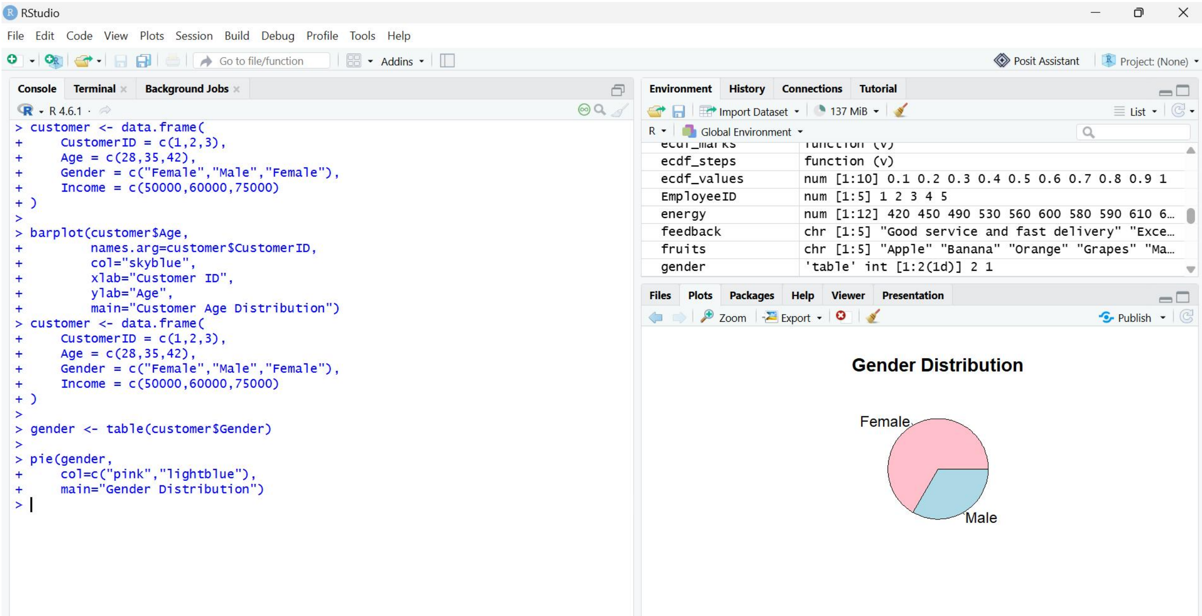
Income = c(50000,60000,75000))

gender <- table(customer$Gender)

pie(gender,

col=c("pink","lightblue"),

main="Gender
Distribution")



Q3. Table

customer <- data.frame(

CustomerID = c(1,2,3),

Age = c(28,35,42),

Gender = c("Female","Male","Female"),

Income = c(50000,60000,75000)

)

Customer

RStudio

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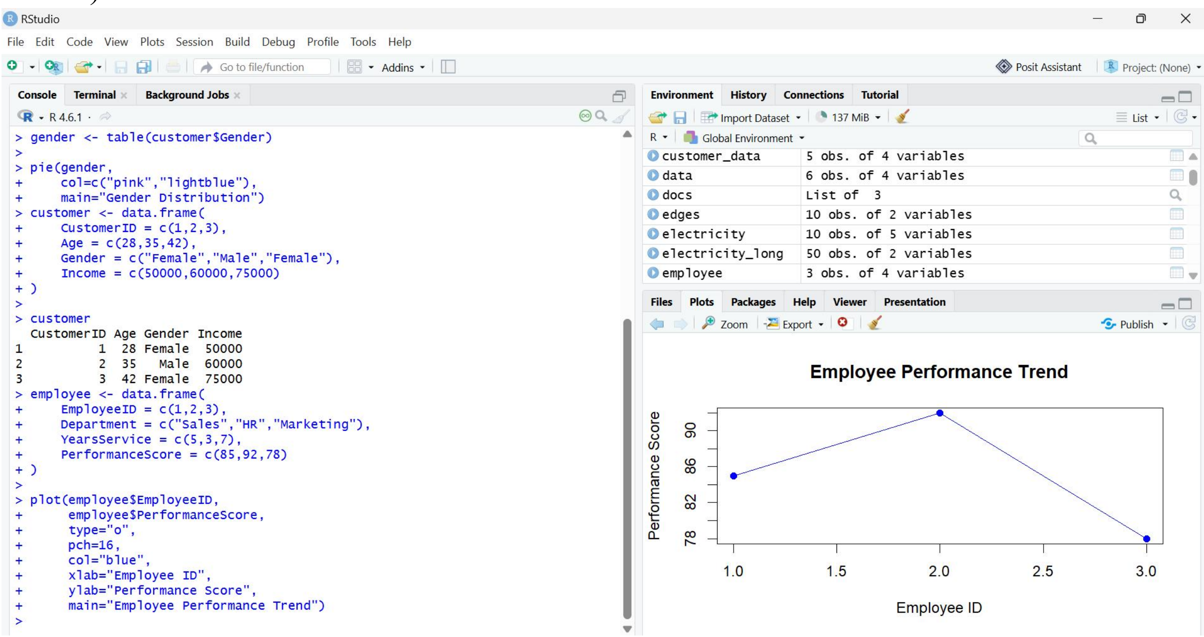
R 4.6.1

```
+ )
>
> barplot(customer$Age,
+         names.arg=customer$CustomerID,
+         col="skyblue",
+         xlab="Customer ID",
+         ylab="Age",
+         main="Customer Age Distribution")
> customer <- data.frame(
+     CustomerID = c(1,2,3),
+     Age = c(28,35,42),
+     Gender = c("Female","Male","Female"),
+     Income = c(50000,60000,75000)
+ )
>
> gender <- table(customer$Gender)
>
> pie(gender,
+     col=c("pink","lightblue"),
+     main="Gender Distribution")
> customer <- data.frame(
+     CustomerID = c(1,2,3),
+     Age = c(28,35,42),
+     Gender = c("Female","Male","Female"),
+     Income = c(50000,60000,75000)
+ )
>
> customer
  CustomerID Age Gender Income
1          1  28 Female  50000
2          2  35   Male  60000
3          3  42 Female  75000
>
```

SET 8 – Employee Performance Analysis

Q1. Line Chart

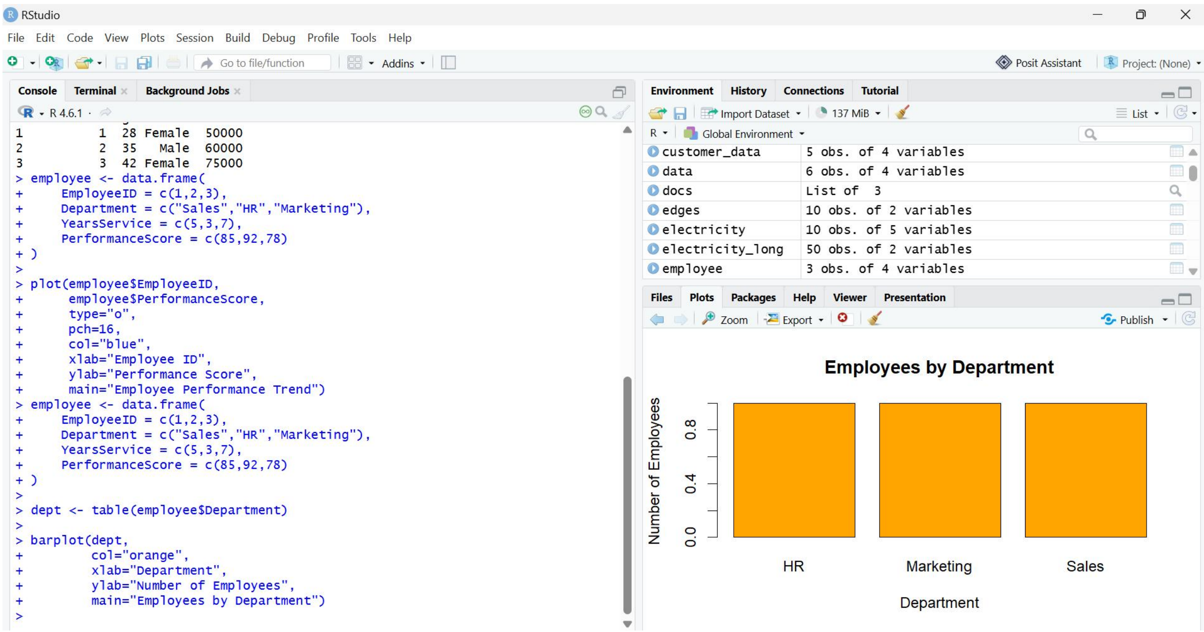
```
employee <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)
plot(employee$EmployeeID,
     employee$PerformanceScore,
     type="o",
     pch=16,
     col="blue",
     xlab="Employee ID",
     ylab="Performance Score",
     main="Employee Performance
Trend")
```



Q2. Bar Chart

```
employee <- data.frame(
```

```
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)
dept <- table(employee$Department)
barplot(dept,
        col="orange",
        xlab="Department",
        ylab="Number of Employees",
        main="Employees by Department")
```



Q3. Table

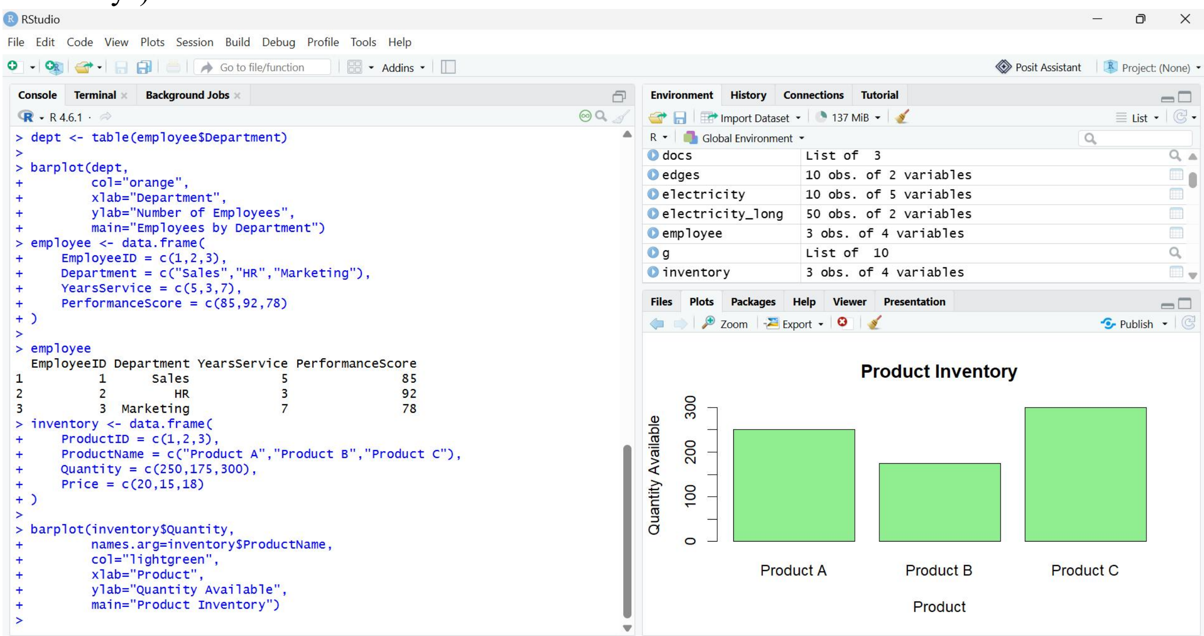
```
employee <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)
Employee
```

```
+ )
>
> dept <- table(employee$Department)
>
> barplot(dept,
+         col="orange",
+         xlab="Department",
+         ylab="Number of Employees",
+         main="Employees by Department")
> employee <- data.frame(
+     EmployeeID = c(1,2,3),
+     Department = c("Sales","HR","Marketing"),
+     YearsService = c(5,3,7),
+     PerformanceScore = c(85,92,78)
+ )
>
> employee
  EmployeeID Department YearsService PerformanceScore
1          1      Sales            5               85
2          2         HR            3               92
3          3  Marketing            7               78
>
```

SET 9 – Product Inventory Management

Q1. Bar Chart

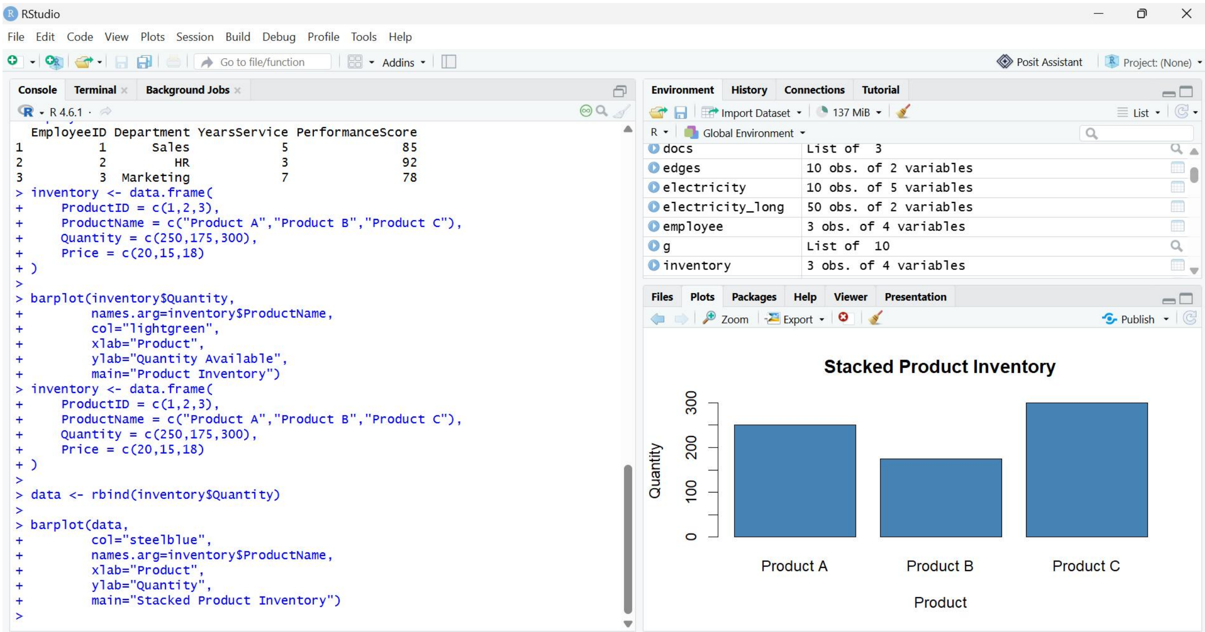
```
inventory <- data.frame(
  ProductID = c(1,2,3),
  ProductName = c("Product A","Product B","Product C"),
  Quantity = c(250,175,300),
  Price = c(20,15,18)
)
barplot(inventory$Quantity,
        names.arg=inventory$ProductName,
        col="lightgreen",
        xlab="Product",
        ylab="Quantity Available",
        main="Product
Inventory")
```



Q2. Stacked Bar Chart

```
inventory <- data.frame(
  ProductID = c(1,2,3),
  ProductName = c("Product A","Product B","Product C"),
```

```
  Quantity = c(250,175,300),
  Price = c(20,15,18)
)
data <- rbind(inventory$Quantity)
barplot(data,
        col="steelblue",
        names.arg=inventory$ProductName,
        xlab="Product",
        ylab="Quantity",
        main="Stacked Product Inventory")
```



Q3. Table

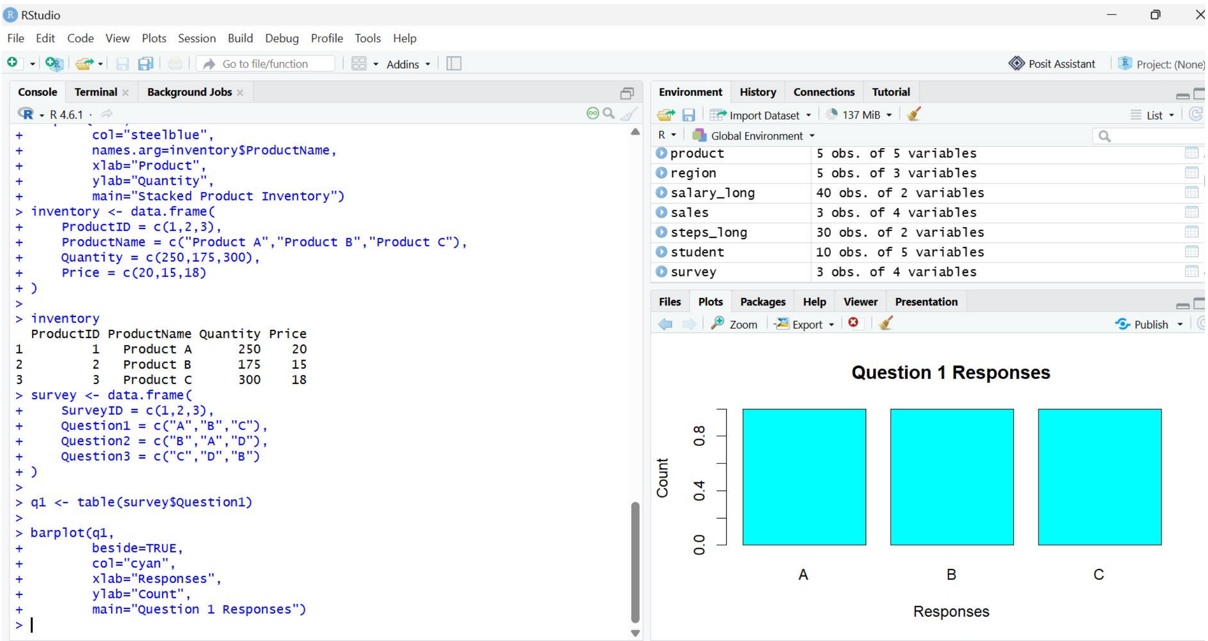
```
inventory <- data.frame(
  ProductID = c(1,2,3),
  ProductName = c("Product A","Product B","Product C"),
  Quantity = c(250,175,300),
  Price = c(20,15,18)
)
Inventory
```

```
> barplot(data,
+         col="steelblue",
+         names.arg=inventory$ProductName,
+         xlab="Product",
+         ylab="Quantity",
+         main="Stacked Product Inventory")
> inventory <- data.frame(
+     ProductID = c(1,2,3),
+     ProductName = c("Product A","Product B","Produ
+     Quantity = c(250,175,300),
+     Price = c(20,15,18)
+ )
>
> inventory
  ProductID ProductName Quantity Price
1         1   Product A      250    20
2         2   Product B      175    15
3         3   Product C      300    18
> 
```

SET 10 – Survey Responses Analysis

Q1. Grouped Bar Chart

```
survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","D"),
  Question3 = c("C","D","B")
)
q1 <- table(survey$Question1)
barplot(q1,
        beside=TRUE,
        col="cyan",
        xlab="Responses",
        ylab="Count",
        main="Question 1 Responses")
```



Q2. Stacked Bar Chart

```
survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
```

Question2 = c("B","A","D"),

Question3 = c("C","D","B")

)

q1 <- table(factor(survey$Question1, levels=c("A","B","C","D")))

q2 <- table(factor(survey$Question2, levels=c("A","B","C","D")))

q3 <- table(factor(survey$Question3, levels=c("A","B","C","D")))

data <- rbind(q1,q2,q3)

barplot(data,

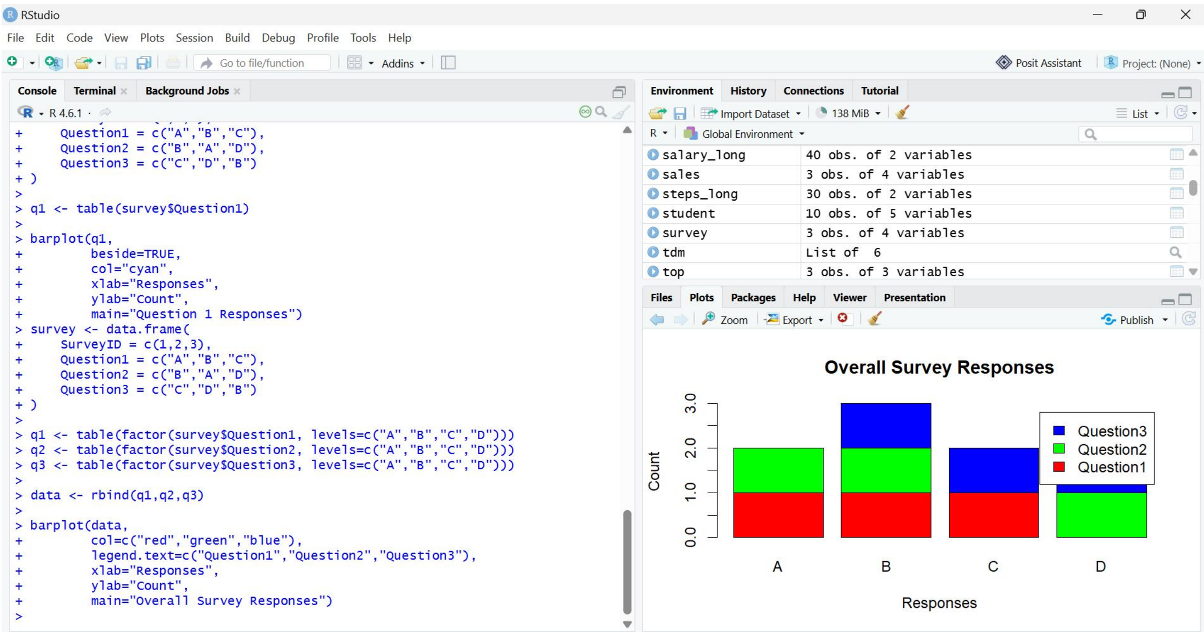
col=c("red","green","blue"),

legend.text=c("Question1","Question2","Question3"),

xlab="Responses",

ylab="Count",

main="Overall Survey Responses")



Q3. Table

survey <- data.frame(

SurveyID = c(1,2,3),

Question1 = c("A","B","C"),

Question2 = c("B","A","D"),

Question3 = c("C","D","B"))

Survey

```
> barplot(data,
+         col=c("red","green","blue"),
+         legend.text=c("Question1","Question2","Question3"),
+         xlab="Responses",
+         ylab="Count",
+         main="Overall Survey Responses")
> survey <- data.frame(
+     SurveyID = c(1,2,3),
+     Question1 = c("A","B","C"),
+     Question2 = c("B","A","D"),
+     Question3 = c("C","D","B")
+ )
>
> survey
  SurveyID Question1 Question2 Question3
1        1         A         B         C
2        2         B         A         D
3        3         C         D         B
>
```